

The empirical recurrence for OEIS A238250: recovering a conjecture that is not written down

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Abstract

OEIS A238250 records “Empirical recurrence of order 81”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 400$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 81, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 81 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A238250 is “Number of $(n+1) \times (3+1)$ 0..1 arrays with 2×2 subblock sums lexicographically nondecreasing columnwise and rowwise.”. It has offset 1 and begins

121, 804, 6828, 43456, 284992, 1473792, 7616082, 32986844, ...

The entry states:

Empirical recurrence of order 81 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:49:01, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 455$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 400$ of the 455, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 400$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 800$ exact terms of the model returns order 81 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{81} c_i a(n-i),$$

c_1	4	c_{22}	−18060943908	c_{43}	−45277405860500	c_{64}	910071726784
c_2	48	c_{23}	−140030373840	c_{44}	79030002738110	c_{65}	−147649110400
c_3	−210	c_{24}	79347451904	c_{45}	43221086320300	c_{66}	−284768617472
c_4	−1099	c_{25}	418883949400	c_{46}	−84436947932680	c_{67}	57176568320
c_5	5350	c_{26}	−285975097632	c_{47}	−36417645347000	c_{68}	75364016896
c_6	15892	c_{27}	−1109214260280	c_{48}	81039869937515	c_{69}	−17801674240
c_7	−88120	c_{28}	876750050276	c_{49}	26802642850300	c_{70}	−16541828096
c_8	−161506	c_{29}	2611021295560	c_{50}	−69792438824464	c_{71}	4447649792
c_9	1054840	c_{30}	−2331902604400	c_{51}	−16923198828674	c_{72}	2930459648
c_{10}	1207264	c_{31}	−5480996160752	c_{52}	53837059591097	c_{73}	−876431360
c_{11}	−9780244	c_{32}	5446257861065	c_{53}	8852633427910	c_{74}	−402612224
c_{12}	−6652562	c_{33}	10283754568300	c_{54}	−37103583130236	c_{75}	131686400
c_{13}	73096940	c_{34}	−11260135151920	c_{55}	−3522953800800	c_{76}	40251392
c_{14}	25088056	c_{35}	−17270747955950	c_{56}	22769277142988	c_{77}	−14213120
c_{15}	−452531800	c_{36}	20724054362435	c_{57}	744826692520	c_{78}	−2605056
c_{16}	−38499923	c_{37}	25977792633850	c_{58}	−12387943673184	c_{79}	983040
c_{17}	2366482580	c_{38}	−34087434088180	c_{59}	289086944160	c_{80}	81920
c_{18}	−286961136	c_{39}	−34987335541400	c_{60}	5942637319696	c_{81}	−32768
c_{19}	−10607568510	c_{40}	50242899311990	c_{61}	−440096653216		
c_{20}	3099148591	c_{41}	42139948635400	c_{62}	−2496184392768		
c_{21}	41215170458	c_{42}	−66477033985120	c_{63}	299946197760		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 81, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $81 + 4$ terms removed returns order 81 as well, so $\deg q_{\min} = 81 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $81 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 81 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A238250.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.